18.0 HUMAN FACTORS ENGINEERING

18.1 CONTROL ROOM DESIGN

18.1.1 CONTROL ROOM DESCRIPTION

18.1.1.1 Introduction

In accordance with the TMI Task Action Plan I.D.1 of NUREG-0660, NUREG-0694, and NUREG-0737 (references 1, 2, and 3, respectively) a control room design review was performed. The purpose of this review was the identification and analysis of human engineering deficiencies (HEDs) so they could be corrected in the final design or accommodated in the administrative and training procedures. This chapter presents the current control room design, the review procedures, and their results.

The preliminary control room design review included an assessment of the following:

- Control room layout.
- The adequacy of the information provided.
- The arrangement and identification of important controls and instrumentation displays.
- The usefulness of the audio and visual alarm systems.
- The information recording and recall capability.
- Other considerations of human factors that have an impact on operating effectiveness.

The final control room design review was conducted prior to licensing, as required by Supplement 1 to NUREG-0737, dated December 17, 1982. This review commenced approximately 1 year prior to Unit 1 fuel loading.

18.1.1.2 Description

The control room plan is shown in figure 18.1-1. The layouts of the Unit 1 main control board (QMCB); electrical auxiliary control board (QEAB); heating, ventilation, and air-conditioning systems panel (QHVC), and miscellaneous systems and equipment panel (QPCP) are shown in drawings 1X3AE01-20, 1X3AE01-21, 1X3AE01-24, 1X5AB01-66, 1X5AB01-67, 1X5AB01-557, 1X5AB01-560, 1X5AB01-563, 1X6AV02-46, 1X6AV02-47, 1X6AV02-48, 1X6AV02-49, 1X6AV02-50, 1X6AV02-60, and AX3D-CA-L50A.

A. QMCB

The QMCB is divided into groups of controls by black demarcation lines. These groups may be entire systems or could be divided into subsystems depending on

the complexity of the system involved. The following are demarcation groups contained on each section of the QMCB:

1. QMCB Section A1

    a. Circulating water.

    b. Turbine plant cooling water.

    c. Turbine plant closed-loop cooling water.

    d. Instrument and service air.

    e. Nuclear service cooling water.

    f. Reactor makeup.

    g. Component cooling water.

    h. Auxiliary component cooling water.

    i. Containment spray system.

    j. Safety injection (SI) pumps.

2. QMCB Section A2

    a. Residual heat removal (RHR).

    b. Chemical and volume control system (CVCS) charging pumps.

    c. Waste processing system.

    d. CVCS letdown.

    e. CVCS seal injection.

    f. CVCS boron injection.

    g. CVCS boron meter.

    h. CVCS thermal regeneration.

3. QMCB Section B1

    a. Main steam isolation valves.

    b. Main steam, steam dump, and steam generator blowdown.

    c. Feedwater, steam generator feed pump turbine A.

    d. Feedwater, steam generator feed pump turbine B.

    e. Auxiliary feedwater.

f. Feedwater and condensate system.

g. Main steam and main steam relief drains.

4. QMCB Section B2

   a. Main steam and main steam relief drains.

   b. Extraction steam.

   c. Turbine.

   d. Generator.

5. QMCB Section C

   a. Reactor control.

   b. Reactor control and nuclear instrumentation.

   c. Reactor coolant system (RCS) pressurizer pressure.

   d. RCS pressurizer level.

   e. RCS reactor coolant pumps.

6. QMCB Section D

   a. Head vents.

   b. Post-accident monitoring system (PAMS) plasma displays.

   c. Main steam isolation valve bypass valves.

B. QEAB

1. QEAB Section A - Plant Auxiliaries

   QEAB section 1A covers electrical mimic boards for Unit 1 systems (common for both units on 1A and 1B). QEAB section 2A covers electrical mimic boards for Unit 2 systems.

2. QEAB Section B

   QEAB section 1B covers electrical mimic boards for Unit 1 system switchyards. QEAB section 2B covers electrical mimic boards for Unit 2 switchyard.

C. QHVC

1. QHVC Section 1

   a. Control building and equipment room, level 4.

   b. Control building auxiliary relay room and computer room exhaust.

   c. Control building control room (normal), level 1.

   d. Control building control room engineered safety features (ESF), level 1.

   e. Equipment room ESF, level B.

   f. Control building electrical penetration room, level B.

g. Electrical tunnel ventilation.

2. QHVC Section 2

a. ESF coolers.

b. Piping penetration area and recycle holdup tanks.

c. Containment heat removal ESF.

d. Containment heat removal (normal).

e. Containment air purification.

f. Containment air purification and post-accident cavity.

g. Containment control rod drive mechanism cavity and reactor support.

h. Diesel generator building.

i. Control building levels A and B (normal) and lighting switchgear (normal).

3. QHVC Section 3

a. Auxiliary building (normal).

b. Auxiliary feedwater pumphouse and control building control room chiller room.

c. Piping penetration ventilation and turbine building exhaust fans.

d. Control building cable spreading rooms, levels A2 and A3.

e. Control building levels 1 and 2 smoke exhaust.

f. Fuel handling building normal exhaust/air-conditioning units and fuel pool area ventilation.

g. Fuel handling building system isolation and post-accident air-conditioning filter units.

h. Fuel handling building normal air-conditioning units supply header isolation.

D. QPCP

1. Section 1

a. Plant demineralized water (Unit 1 only).

b. Nuclear service cooling tower makeup water.

c. River makeup water (Unit 1 only).

d. Waste water.

e. Turbine plant cooling water.

f. Core monitor panel.

g. Main steam atmospheric relief valve.

h. Cooling water system temperature, train A.

i. Cooling water system temperature, train B.

2. QPCP Section 2

a. SI test line valves.

b. Main steam.

c. Steam generator feed pump turbine drains.

d. Reheater drains.

e. Auxiliary feedwater.

f. Feedwater and condensate chemical injection.

g. Containment building and auxiliary building drains.

h. Fire protection.

i. Refueling water storage tank sludge mixing.

3. QPCP Section 3

a. Train A condensate and feedwater.

b. Train A main steam.

c. Train A steam generator blowdown sampling.

d. Train A fire protection and containment service air.

e. Train A line break monitors and CVCS letdown header.

f. Train A containment hydrogen monitoring.

g. Train A containment air monitoring.

h. Train A pressurizer liquid.

i. Train A SI sample.

j. Train B condensate and feedwater.

k. Train B main steam.

l. Train B steam generator blowdown sampling.

m. Train B fire protection and containment service air.

n. Train B line break monitors and CVCS letdown header.

o. Train B containment hydrogen monitoring.

p. Train B containment air monitoring.

q. Train B pressurizer liquid.

r. Train B SI sample.

These control panels and their components provide display, control, alarm, and computer functions to assist in remote control, monitoring, and recording of plant operations during startup, normal, shutdown, and emergency operations. The control room and its boards are designed and configured to enable the operator to control the plant as required in an efficient manner without undue risk to public health and safety or potential problems to plant equipment. Should the need exist, the operator can take appropriate measures or can initiate protective action using the control room controls.

If the control room becomes uninhabitable, the necessary controls and displays for shutdown of the plant are provided at two secure, remotely located shutdown panels. (See section 7.4.)

18.1.2 DESIGN BASES

The following design bases constitute the design requirements for the control room evaluation.

18.1.2.1 Anthropometrics

This is defined as statistical data of physical dimensions for the control room personnel.

A. The QMCB is a combination bench-vertical type.

B. The remainder of the control boards are the vertical type.

18.1.2.2 Auditory Communications

This is defined as indications or displays for equipment or subsystem conditions as interpreted by the control room personnel in auditory modes.

A. The annunciator is designed to provide five mutually discriminable audible signals to allow the operator to be immediately cognizant of an alarm and to be able to distinguish the relative board location of that alarm based on sound characteristics. To ensure immediate notice by the operator, each audible alarm is designed to have a minimum sound level differential of 10 dBA above the ambient level.

1. The QMCB annunciator panels have a medium-pitched, medium-warble horn sound.
2. The QEAB has a low-pitched, low-warble horn sound.
3. The QHVC is designed to have its own horn with a low-pitched, high-warble sound.
4. The QPCP is designed to have a separate horn with a high-pitched, high warble sound.
5. The ring-back (reset of alarm) is common to all control boards, using a bell tone.
6. First-out panel on the QMCB has a high-pitch, high-warble sound (notably different than the QPCP).

18.1.2.3 Plant Communications

This is defined as the means provided for intercommunication between the control room, technical support center, emergency operations facility, and personnel on duty in various parts of the plant, as described below.

A. Communication to/from the control room is provided using sound-powered telephone systems. The equipment is available for maintenance (six jacks/box), refueling (two jacks/box), and shutdown purposes (two jacks/box).

B. The communication system within the plant is a public address system (one page line and party line common to all zones and four party lines). This system is for roving operator communications and is provided with the capability for the merger of four zones: Unit 1, Unit 2, administration building, and outside areas. This system provides remote merging in:

1. Unit 1 control room.
2. Unit 2 control room.
3. Central alarm station.
4. Secondary alarm station.
5. Captain's office in the PESB.

This also functions as an emergency alarm (supplied with emergency power from an uninterruptible power supply system). This system is actuated by a multi-tone generator applicable for discriminable tones. Capability for initiation and selection of the tone for the emergency alarm is provided in the control room.

C. Plant offsite communication system uses the public telephone systems.

D. Other communication facilities are:

1. Microwave communications at 57 MHz.
2. Walkie-talkie with external antenna for radio contact between the control room, technical support center, emergency operations facility, and field personnel.
3. Security intercom linking control room, security subsystem, and security monitoring station.

4. Federal Telephone System (FTS) for the emergency notification system (ENS) and the health physics network.

Communications are further discussed in subsection 9.5.2.

18.1.2.4 Controls

Controls are defined as devices actuated by the control room personnel (manual mode) or by the systems (automatic mode) to influence system functioning.

A. The control devices used on the instrument panels are designed to minimize the size of the panels, while still allowing convenient and reliable operation. Insofar as is practicable, uniform types of control devices are used throughout the control room.

B. Control devices of redundant trains are designed for separation by physical barriers where there is an operational requirement for redundant trains to be close to each other.

C. Control devices for safety systems are designed to satisfy Seismic Category 1 requirements.

D. Control switches selected for the auxiliary board and panels are similar to those selected for the QMCBs (Electroswitch series 20 or equivalent).

E. Control switch development is designed to comply with the following:

1.	Left-right (right throw) or bottom-top (up-throw)	Start, close (circuit breaker), open (valve), on, raise, and other positive or increasing functions and associated indicating lights.
2.	Right-left (left throw) or top-bottom (down-throw)	Stop, trip (circuit breaker), close (valve), off, lower, and other negative or decreasing functions and associated indicating lights.

F. Where possible, automatic controls for parallel valves are designed to be kept in parallel (left to right). The series valves are designed to be kept in series (top to bottom) to make it compatible with the flow direction on the control board layout. Flow paths for system controls are designed for layout on the QMCB starting with suction, pump, and discharge going from top to bottom.

G. The analog control stations are Westinghouse 7300 series operator interface modules, which use Westinghouse-qualified designs for split architecture. Controls on the operator interface modules are marked to indicate in which direction to operate the controller to increase or decrease the variable to avoid operator errors.

H. Controls are designed for functional arrangement on control boards so that systems or equipment supporting or interacting with the desired function are located in the same general area on the control board for ease of operation.

I. The controls associated with core reactivity are located in the central area of the QMCB (section C) along with the other principal reactor controls.

J. Speed and flow controls for the charging pumps are located so that the flow control and pressurizer control are in the reactor control section of the QMCB (section C).

K. Controls for heating, ventilation, and air-conditioning (HVAC) and control rod drive motor fans are on the QHVC panel.

L. Control switches for rod motion are designed to have a clear line of vision to the rod step counters and count rate recorder so that the operator can continuously observe the effects of the rod motion.

M. Control switches for reactor trip and manual SI are located away from the other control switches to avoid inadvertent actuation.

N. Functional groups of controls are located in accordance with operational sequence having left-to-right order and/or top-to-bottom order of use based on the accepted operator compatibility principles, namely, spatial, movement, and conceptual (e.g., suction isolation valve, pump, and discharge isolation valve, in that order).

18.1.2.5 Control and Display Integration

This is defined as the relative locations of related control and display devices.

A. Controls and displays (including annunciators and recorders) are arranged in functional groups and in a functional sequence with the displays above their corresponding controls to prevent the arm of the operator from obscuring the indicated response to control adjustments.

B. Main feedwater and bypass feedwater valve controls and displays are located close to each other to facilitate transferring modes of operation as power level is changed.

C. Frequently used control and display systems are located on or near the central portion of the QMCBs to minimize operator movement.

18.1.2.6 Design for Maintainability

This is defined as the degree to which panels are designed to accommodate preventive maintenance and corrective repair or replacement functions as quickly as possible with minimum disturbance to operator functions.

A. Control boards in the control room are designed with split architecture controllers (controllers separate from process module) to facilitate maintenance and to reduce congestion in the operator's area of the control room.

B. Control panel components are primarily modular equipment to simplify maintenance as well as to reduce the maintenance time.

C. Maintenance accessibility is designed for minimum operator maintenance and technician interference by maintaining sufficient aisle clearance and by positioning panels so that there is adequate aisle space for panel back access.

18.1.2.7 Design for Personnel Requirements

This is defined as the norms in the power industry for control room personnel physical and mental capabilities.

A. A shift supervisor and operators are available in the control room for each shift as specified in Technical Specification 5.2.2, 10 CFR 50.54, and the Technical Requirements Manual.

B. A shift technical advisor or similarly qualified senior reactor operator is available to assist in the interpretation of information and control actions.

C. The qualifications for the operators are as specified in chapter 13.

18.1.2.8 Environment

This is defined as the external physical elements affecting the control room personnel in their performance of their functions. The environment for the main control room is designed for the following conditions:

A. Temperature of the control room is normally 75°±5°F with maximum allowable temperature of 85°F.

B. Pressure of the control room is slightly above atmospheric.

C. Relative humidity is maintained between 10 and 60 percent.

D. The fluorescent lighting in the control room is designed to provide a uniform horizontal intensity of 50 to 100 fc adjustable in the control room general area, 50 to 100 fc adjustable at the console surface, and approximately 30 fc during an emergency resulting from loss of normal 120-V ac lighting. The control room lighting system is further described in subsection 9.5.3.

E. Acoustical provisions are included in the design of the primary operating area of the control room to minimize the ambient noise to a sound level of 50 to 65 dBA.

18.1.2.9 Hazards/Safety

These are defined as the physical, chemical, and other hazards to control room personnel.

A. The HVAC system is designed to provide automatic protection against radiation and smoke. Manual operation provides protection against toxic chemicals.

B. The control room design includes protection against dangerous voltage and ground faults by appropriate use of fuses, breakers, and grounding techniques.

C. The control room is designed for operator warning should a hazardous situation arise.

D. The control room is provided, where required, with special ladders for bulb replacement.

E. Switch handles and other devices are designed so as not to protrude from boards and panels in a manner that could produce accidental actuation. A guard rail around the QMCB also prevents accidental activation.

F. The control room is provided with fire, smoke, and radiation detection. The fire hazard and protection analysis of the control room is provided in appendix 9A.

G. The control room is designed for automatic warnings against unsafe or emergency conditions in the plant by alarms.

H. Automatic actuation for control room isolation is also provided with operator actuation as a backup.

I. Accessibility of emergency and protective equipment to the control room operators is discussed in section 6.4.

J. Control room suspended ceiling is designed and constructed to ensure that the ceiling will not fall or compromise the functioning of safety-related equipment during or after a safe shutdown earthquake.

18.1.2.10 Information Utilization

This is defined as the information processing facility, such as entry, access, and retrieval, provided to improve control room personnel performance. The control room and boards are designed to have the following features:

A. Automated control of safety feature equipment if action is required within 10 min from the time indication is available for showing that action is required.

B. Operator involvement in selection of automatic versus manual control of loops.

C. Selection of Monitors and other computer display functions.

D. Operator involvement in:

1. Startup.
2. Shutdown.
3. Load change.
4. Base load.
5. Trips.
6. Accidents.
7. Emergencies (including fire and security).

E. Selection of the particular train to use when a system has two trains.

F. Display, diagram, or information selection (including details of annunciated items if required).

G. Data handling and display (calculation, evaluation, and screening information before display).

H. Annunciator interpretation, alarm handling, and display.

I. Manual overrides for essential system.

J. Backup facility for highly significant parameter control.

K. Display of important parameters with the ability to call up backup information as needed by the operator to take any required action.

18.1.2.11 Labeling (Grouping, Marking)

This is defined as alphanumeric, color, and other visual methods used for controls and displays to improve the performance of control room personnel.

A color coding scheme for the switchplates on the QMCB, QEAB, QHVC, QPCP, PSDA, and PSDB is implemented in a plant procedure.

Additionally:

A. Controls located on the bench board section of the QMCB are grouped by subsystems divided by the demarcation lines (paragraph 18.1.1.2.A) and enveloped, where room permits, with hierarchical labels for subsystems at the top center of the envelopes to improve recognition of functional grouping.

B. Labels are located to minimize interference with operator view and to avoid interference with other control functions.

C. Labels are designed for legibility and visibility based on the contrast between the lettering and its background.

D. Labels are designed to have white letters on black tags. Black on white is sometimes used to highlight a display.

E. Labels are designed with 3/16-in. capital letters. Annunciator window engravings are designed with 1/4-in. or 3/16-in. letters. The 1/4-in. letters are used to improve readability when message length permits.

F. Controls and displays are labeled with service description and tag number engravings. In addition, the control switch modules shall contain engravings for switching development functions, equipment-actuated functions, and train, if applicable.

G. Labels are placed above the instruments and on the escutcheon plates for control switches to allow adequate viewing from a distance of about 3 ft.

H. Labels for similar devices throughout a board are designed to be uniform in style, size, lettering, and use of abbreviations, with the exception of integral panels supplied by the vendor and inserted into the boards or panels as a unit.

I. Labels are designed and mounted so that they cannot easily be damaged or removed.

J. Labels and tag numbers are designed for accessibility and visibility during maintenance.

K. Labels are concise with minimum repetitive information and are directly usable with minimum decoding and interpretation of the service descriptions and abbreviations. A hierarchical label is used to highlight functional grouping.

L. Labels are not designed to describe engineering characteristics, name of manufacturer, trademarks, or nonfunction-related nomenclatures of the equipment.

M. Shades of colors used for mimics on the QEABs are designed to have maximum contrast between the mimic bus and the boards.

N. Safety-related post-accident monitoring instrumentation is identified by a dark red line on the black bezel base of each instrument.

18.1.2.12 Visual Display (Meters, Recorders, Lights, Monitors, and Annunciators)

This is defined as indication and display of equipment or subsystem conditions in visual modes.

18.1.2.12.1 General

A. Where possible, displays are arranged in the sequence in which they are used.

B. Unusual aids, such as stools and ladders, extra lighting, etc., are not required to read or gain access to a display.

C. Information for different types of activation are not combined unless activation requires the same information.

D. In a standing position, the most frequently used displays are, insofar as possible, located at the eye level of the operator.

E. Displays frequently used in conjunction are grouped together.

F. Displays are located where they can be read with accuracy and minimum parallax.

G. Scale face and graduation markings are designed to have a high degree of contrast.

H. Displays are designed so that glare does not interfere with readability when viewed from a reasonable location.

I. Displays which cannot or may not be watched continuously, but which need vigilance in monitoring, are designed to have a suitable auditory or visual alarm.

J. Multiple displays grouped together are designed to have uniform brightness across the full range of the display faces.

18.1.2.12.2 Meters

A. Meters, dials, and instruments are designed for size and location so they can be read from normal operating positions.

B. Information presentations are designed in such a form that minimum interpretation or decoding is necessary for operator implementation.

C. The meter pointer is designed to extend to, but not obscure, the graduation marks.

D. The meter pointer is designed for mounting as close as possible to the dial face to reduce or eliminate parallax and shadows.

E. Meter scales generally have graduations of 1, 2, 5, 10, or any of their multiples.

F. The display indicator does not move after the control movement stops.

G. System indicators in a flow path are arranged in accordance with flow path from left to right; i.e., inlet temperature, flow, outlet temperature.

H. Meters have nonglare covers where needed.

I. The meter scales of selected instruments are color coded to indicate the normal operating region.

J. T_{avg}/T_{ref} indicators are located next to each other for easy comparisons.

18.1.2.12.3 Recorders

A. Whenever possible, small continuous multipen recorders, having dimensions of approximately 6 in. by 6 in. or 3 in. by 6 in. are used on the QMCB.

B. Scanning-type indicators are of the low-profile type to reduce panel dimensional requirements.

18.1.2.12.4 Lights

A. Indicating lights are designed to show information regarding equipment status.

B. Indicating lights are designed to show status derived from equipment response and not just to indicate control positions. They are designed to show information needed for effective system operation.

C. Both the red and the green lights are designed to be lit for motor-operated valves and other devices with considerable travel time while the device is in an intermediate position.

D. The indicating lights for different operational conditions are designed as follows:

1. Red - Equipment or process operating, flowing, or in an increasing condition, breaker closed, valve open.
2. Green - Equipment or process not operating, not flowing, or in a decreasing condition, breaker open, valve closed.
3. Amber - Trip (automatic protection and not by operator action or process; exceptions exist for a few valve indications where it is necessary to heighten operator awareness of off-normal valve configurations).
4. White - Electrical potential available or other special uses.

Use of blue indicating lights is limited because of the low level of brightness obtained with this color.

E. Indicating lights are designed for showing the status of equipment interlocked with the controls of associated equipment.

F. Indicating lights installed on the control boards are designed for uniform size and use standard abbreviations for their engravings.

18.1.2.12.5 System Status Monitoring Panel

The system status monitoring panel (QBPS), shown in figure 18.1-1, is described in detail in subsection 7.5.5. All lights on the QBPS are white.

18.1.2.12.6 Annunciators

A. The annunciator displays alarms that denote an abnormal equipment status or plant condition. Nuisance alarms are kept to a minimum by suitable permissive interlocking contacts.

B. Annunciators are designed for high reliability, using solid state equipment.

C. The annunciator system is designed to provide reasonable assurance that failure of one alarm circuit does not disable any other alarm circuit. Preoperational testing verified that failure of one alarm card did not disable the adjacent alarm circuit. This was demonstrated by removing an alarm card and testing the adjacent circuit.

D. Annunciator and computer alarms are designed not to be duplicated, except for the QBPS and selected computer sequential alarms.

E. Each section of the QMCB or QEAB has a three-pushbutton station to acknowledge, test, and reset the annunciator windows located in that section of the board.

F. The HVAC panel and miscellaneous system/equipment panel have the capability to acknowledge, test, and reset the annunciator only by the pushbutton station located on their respective panels.

G. All annunciators are designed as non-Class 1E. The annunciator contacts accept signals from Class 1E circuits which are isolated from the Class 1E system.

H. The power supply for annunciators is obtained from a non-Class 1E battery.

I. Control room annunciator windows are designed for white opaque; colors are obtained by installing color lenses.

J. The alarm sequences for the main control room annunciator are designed as indicated in tables 18.1-1, 18.1-2, and 18.1-3.

K. The three-pushbutton stations have the following functions:

1. Acknowledge - To silence the horn and put the alarm light on steady.
2. Test - To sound the horn and put all alarm lights on a panel or control board section on fast (or gallop for first-out sequence alarms) flash.
3. Reset - To silence the horn and extinguish the lamp.

L. Annunciator nominal window dimensions are 1.4 in. by 3 in.; letter size is 3/16 in. or 1/4-in.

M. Dual alarms (such as high-low, high-high, etc.) are only allocated a single alarm window, provided a second point of information is available to distinguish between the two conditions (such as an indicator, monitor lights, or recorder).

N. Each local annunciator panel is designed to have a common retransmitting contact to operate a single window in the control room annunciator (exceptions are the emergency diesel generator local annunciator panels which are duplicated on the QEABs).

O. Reflash (occurrence of another alarm on the same window after acknowledge) capability is provided in special cases for annunciator windows with multiple contacts if required for safe and reliable operation of the plant.

18.1.2.12.7 Monitors

These are defined as display devices with television-type screens to indicate a comprehensive group of information and with flexibility to call up any stored information as required.

A. The control board is designed to locate a Monitor on the display section of the QMCB board section C for displaying the annunciated parameters from the plant computer.

B. The control board is designed to locate a Monitor at the normal eye level on the display section of the QMCB section B2 for displaying parameters from the plant computer.

C. The control room is designed to mount one Monitor on the operator desks to provide graphic displays.

D. The control room is designed to mount two free-standing safety parameter display system Monitor consoles located near the QMCB, as shown in figure 18.1-1.

E. The vertical section of control board QMCB Section B2 is designed to locate three monitors along with keyboards and touch pads for providing operator interfaces to control the main turbine, feed pump turbines, and the generator excitation system (Unit 1 only).

18.1.2.12.8 Plasma Display Modules

Two redundant dot matrix-type graphic/alphanumeric displays shall be provided in section D of the QMCB. These modules shall display the process variables classified in Regulatory Guide 1.97, Revision 2, as Category 1, as well as other selected variables. Plasma displays are a part of the PAMS described in section 7.5. They interface with the redundant display processing units and meet the single failure requirements.

18.1.2.13 Workspace

This is defined as the space surrounding the area designated "at the controls."

A. The control room area is designed to include dual control rooms (one per unit) with a supervisor's area positioned to observe both central control areas.

1. Arrangements of the most frequently used boards and displays are designed for location readily visible to the operator in the central control area. Control boards are designed to be arranged to provide the best accessibility; i.e., shortest walking distance.

2. Layout designs of the QMCB, QEAB, desks, and other panels are arranged so that physical interference among operators working in the central control area is minimized.

3. The design for the central control area space and aisles allows efficient movement of operators between areas during normal, emergency, startup, and shutdown operations.

4. The lines of sight to a display are designed so as not to be obstructed by poor arrangement of equipment.

5. A restricted area between the operators' desks and the QMCB and QEAB is designed for the use of licensed operators only.

18.1.2.14 References

1. U.S. Nuclear Regulatory Commission, "NRC Action Plan Developed as a Result of the TM1-2 Accident," NUREG-0660, May 1980.
2. U.S. Nuclear Regulatory Commission, "TMI Related Requirements for New Operating Licenses," NUREG-0694, June 1980.
3. U.S. Nuclear Regulatory Commission, "Clarification of TMI Action Plan Requirements," NUREG-0737, November 1980.

TABLE 18.1-1

MAIN CONTROL ROOM ANNUNCIATOR ALARM SEQUENCE
FIRST-OUT SEQUENCE OF OPERATION (MODIFIED)[a]

Condition		Field Contact	Lamp	Horn 1	Horn 2
Normal		Normal	Off	Off	Off
Alarm	First	Abnormal	Gallop flash	On	Off
	Subsequent	Abnormal	Fast flash	On	Off
Return to normal before acknowledge	First	Normal	Gallop flash	On	Off
	Subsequent	Normal	Fast flash	On	Off
Acknowledge	First	Abnormal	Slow flash	Off	Off
	Subsequent	Abnormal	Steady	Off	Off
Return to normal	First	Normal	Slow flash	Off	On
	Subsequent	Normal	Slow flash	Off	On
Reset	First	Normal	Off	Off	Off
	Subsequent	Normal	Off	Off	Off
Test	First	Normal	Gallop flash	On	Off

a. All windows will be on a first-out condition during the test sequence.

TABLE 18.1-2

MAIN CONTROL ROOM ANNUNCIATOR ALARM SEQUENCE
RING-BACK SEQUENCE OF OPERATION

Condition	Field Contact	Lamp	Horn 1	Horn 2
Normal	Normal	Off	Off	Off
Alarm	Abnormal	Fast flash	On	Off
Return to normal before acknowledge	Normal	Fast flash	On	Off
Acknowledge	Abnormal	Steady	Off	Off
Return to normal	Normal	Slow flash	Off	On
Reset	Normal	Off	Off	Off
Test	Normal	Fast flash	On	Off

TABLE 18.1-3 (SHEET 1 OF 2)

MAIN CONTROL ROOM ANNUNCIATOR ALARM SEQUENCE

Reflash Sequence of Operation - One Incoming Alarm Which Returns to Normal Before Being Acknowledged (Common Window)

Condition	Field Contact	Lamp	Horn 1	Horn 2
Normal	Normal	Off	Off	Off
Alarm[(a)]	Abnormal	Fast flash	On	Off
Return to norma before acknowledge	Normal	Fast flash	On	Off
Acknowledge	Normal	Slow flash	Off	On
Reset	Normal	Off	Off	Off

Reflash Sequence of Operation - Two or More Incoming Alarms (Common Window)

Condition	Field Contact	Lamp	Horn 1	Horn 2
Normal	Normal	Off	Off	Off
Alarm 1[(a)]	Abnormal	Fast flash	On	Off
Alarm 2[(a)]	Abnormal	Fast flash	On	Off
Acknowledge	Abnormal	Steady	Off	Off
Alarm 3[(a)]	Abnormal	Fast flash	On	Off
Alarm 1 returns to normal	Abnormal	Fast flash	On	Off
Acknowledge	Abnormal	Steady	Off	Off
Alarm 2 returns to normal	Abnormal	Steady	Off	Off

TABLE 18.1-3 (SHEET 2 OF 2)

Condition	Field Contact	Lamp	Horn 1	Horn 2
Alarm 3 returns to normal	Normal	Slow flash	Off	On
Reset	Normal	Off	Off	Off

a. Incoming alarm shall always have precedence over return to normal.

18.2 PRELIMINARY REVIEW OF THE CONTROL ROOM AND CONTROL BOARD DESIGN

18.2.1 REVIEW PROCEDURES

18.2.1.1 Introduction

The preliminary review of the control room and control board design was conducted by the General Physics Corporation, with the primary objective of providing an adequate evaluation of the human factors.

The review included primarily those panels which comprised the inner ring of panels in the control room.

Certain additional panels which are not in the inner ring were also reviewed. A listing of the panels reviewed is given below:

Inner Ring Panels	Outer Ring Panels
QMCB A1	HVC
QMCB A2	QPCP
QMCB C	
QMCB B1	
QMCB B2	
QMCB D	
QEAB	

Figure 18.1-1 illustrates the layout of the panels in the main control room.

The review procedures and results are presented in detail in reference 1 and summarized in this section. The review was performed in accordance with the guidelines of NUREG/CR-1580.(2)

The use of engineering checklists provided standards of assessment of various properties of the control room including, among other aspects:

- Functional grouping.
- Anthropometrics.
- Readability of labels.
- Controls discrimination.

These checklists were developed based upon the latest established and recommended human factors engineering criteria.

Several techniques were employed in the review of the VEGP control room. Operators were used to determine how the VEGP control board stood in relationship to similar vintage control boards. Human factors specialists judged the control board against applicable guidelines. Engineering checklists were employed to evaluate individual controls and control/display layout. Scenario evaluations were performed to assess the operability of the board. Using the above methods, the human factors engineering group was able to make assessments of:

- Anthropometrics and panel elements.
- System layout.
- Component usage.
- Readability of displays and labels.
- Coding/discrimination methods.
- Environmental conditions.
- Operability of control board.

Below is a detailed description of methodologies used in anthropometrics and panel elements analysis, system evaluation, component evaluation, and scenario evaluation. The information gathered by use of these human factors methodologies served as a basis for the development of preliminary recommendations. The evaluation of these recommendations and their resolutions are provided in subsection 18.2.2.

18.2.1.2 Anthropometrics

Anthropometric evaluation was performed by comparing the control room design to established human factors guidelines. The main references used were Bechtel drawings, NUREG-1580,[(2)] and MIL-STD-1472B.[(3)]

The guidelines were applied three ways: by scale drawings, by use of a 1/4-in. scale panel silhouette and mannequins scaled for the 95th and 5th percentile male, and by actual measurements of reach and heights taken from the mockup.

Items examined were:

- Panel height.
- Distance of reach.
- Viewing distance.
- Viewing angle.
- Placement of controls and displays in relationship to anthropometric value.

18.2.1.3 System Evaluation

System evaluation was performed after developing engineering checklists from the current applicable human factors engineering guidelines, specifically NUREG/CR-1580.[(2)]

A separate set of questions was asked about each control and each display. A list of these questions is provided below. The questions were asked about each individual component with answers recorded as a yes or no.

A. Control Checklist

1. Are controls placed such that they may be easily operated?
2. Is control located to prevent inadvertent operation?
3. Does control give the information needed for operation of the equipment/system?
4. Is the control clearly marked as to what it does?
5. Do controls move in the culturally normal direction (clockwise for on)?
6. Is this control the best type of control for the function needed?
7. Are controls grouped in a consistent left-to-right, top-to-bottom order?
8. Is control not prone to be misread or its position misinterpreted?

B. Display Checklist

1. Is display prone to be misread or misinterpreted?
2. Is display the best type for indicating system/equipment function?
3. Does display give the type of information needed?
4. Does display movement correspond to the control movement?
5. Is display in reasonable proximity to the control?

18.2.1.4 Component Evaluation

Component evaluation was performed by reviewing all pertinent human factors guidelines, then developing a list of these guidelines applicable to components in the control room. Additions to this list draw heavily on Nuclear Regulatory Commission (NRC) Safety Evaluation Reports.[4][5][6]

These documents were reviewed:

- Electrical one-line diagrams.
- Piping and instrumentation diagrams.
- Control logic diagrams.
- Instrument specification sheets.
- Instrument index.
- Digital rod position indicator specification.
- Specification for the electrical auxiliary board and the miscellaneous control board.
- Various vendor documents were utilized.

Where possible, a hands-on evaluation of controls was performed. This entailed the use of testing devices such as a snap-on torque meter or an Amtex torsional testing machine. In addition, observation of equipment equivalent to that found in the VEGP control rooms was made at various operating simulators. In particular, the Sequoyah and McGuire simulators were used.

Safety evaluation reports were reviewed to determine how the NRC implemented the various guidelines.

18.2.1.5 Scenario Evaluation

Scenario walkthroughs were performed at the control room mockup for Unit 1 to gain an operational perspective on the proposed layout and to evaluate it from the operator's point of view. The primary purpose of this methodology was to assess the logical sequencing of all actions and the flow of required motions to perform the required actions in a timely manner.

This evaluation was performed in three phases. The first phase was the creation of operating procedures from Westinghouse generic procedures. The second phase was the actual walkthrough. The third phase was the scenario evaluation.

Nine scenarios were selected from Westinghouse generic procedures. Four nonprocedural scenarios were added during the review. The nine selected were chosen because they represent a wide variety of plant activities and contain the basic parameters for many more operating tasks. The four shorter scenarios were added to provide a review of observational tasks. The nine scenarios were:

- Immediate action and diagnostics.
- Steam generator tube rupture.
- Operation with natural circulation.
- Plant shutdown from minimum load to cold shutdown.
- Reactor coolant system leak.
- Station blackout.
- Plant startup from cold shutdown to minimum load.
- Loss of reactor coolant.
- Reactor trip.

The four additional scenarios performed were:

- Loss of secondary coolant.
- Starting the diesel generators.
- Controlling steam generator level.
- Changeover from recirculating mode to injection mode.

The information collected by use of these human factors methodologies served as a basis for the development of preliminary recommendations.

The evaluation of these recommendations is provided in subsection 18.2.2.

18.2.1.6 Reviewer Experience

General Physics Corporation, the prime control board reviewer, has had the following experience in design review:

- North Anna, Unit 2 Control Room Review.
- EPRI-RP769, Performance Measurement System for Training Simulators.
- Babcock & Wilcox and U.S. Department of Energy, Disturbance Analysis and Surveillance System.
- Clinch River Breeder Reactor Control Room Review.
- Edison Electric Institute, Operator Selection Study.
- Operability Assessment of Prototype Large Breeder Reactor Designs.
- Electric Power Research Institute, Survey and Analysis of Communication Problems in Nuclear Power Plants.

18.2.2 CONTROL ROOM AND CONTROL BOARD EVALUATION RESULTS

18.2.2.1 Human Factors Deficiencies Categories

The human factor deficiencies were categorized into five general groupings:

A. Administrative, where no hardware solutions were available.

B. Components, where the type of display, means and methods of manipulation, and elimination of the chances for inadvertant or accidental misoperation were considered.

C. Labeling, where the identifications of systems, trains, and devices were reviewed so that quick and easy identification by the operators would be enhanced.

D. Rearrangement of components, where component functional relationships were considered and their actual arrangement on the board was then critiqued.

E. Workshop/environment, where noise, temperature, lighting, and workspace definition were considered which would lead to increased operation effectiveness and reduced distraction; other deficiencies which would negatively affect operator performance.

18.2.2.2 Findings and the Resolutions Thereof

The preliminary control room design review identified human engineering deficiencies at an early stage in the design in order that they could be incorporated in design improvements, operator training, and plant operating and administrative procedures. When final NRC guidance was issued, the review process was repeated in the detailed control room design review, detailed in section 18.3.

18.2.2.3 References

1. General Physics Corporation, "Report on Human Factors Evaluation of Alvin W. Vogtle Nuclear Power Plant Control Room," GP-R-23003, Columbia, Maryland.

2. U.S. Nuclear Regulatory Commission, "Human Engineering Guide to Control Room Evaluation - Draft Report," NUREG/CR-1580, July 1980.

3. U.S. Department of Defense, "Human Engineering Design Criteria for Military Systems, Equipment, and Facilities," MIL-STD-1472B, Washington, D.C., May 1978.

4. U.S. Nuclear Regulatory Commission, "Safety Evaluation Report, Supplement No. 1, Docket Nos. 50-327 and 50-328, Sequoyah Nuclear Plant Units 1 and 2," NUREG-0011, February 1980.

5. U.S. Nuclear Regulatory Commission, "Safety Evaluation Report, Supplement No. 10, Docket No. 50-339, North Anna Power Station Unit 2," NUREG-0053, July 1980.

6. U.S. Nuclear Regulatory Commission, "Safety Evaluation Report, Supplement No. 4, Docket No. 50-311, Salem Nuclear Generating Station Unit 2, NUREG-0512, January 1979.

18.3 DETAILED CONTROL ROOM DESIGN REVIEW

18.3.1 BACKGROUND

Guidance for the control room design review (CRDR) was provided by various NUREGs and Regulatory Guides. A Nuclear Utility Task Action Committee (NUTAC) with staff support from the Institute of Nuclear Power Operations (INPO) developed a generic control room design review implementation plan from these guidelines. NRC guidance and NUTAC guidelines were used by Georgia Power Company (GPC) for the development of the detailed CRDR process.

The detailed control room design review stands alone in documenting compliance with NRC requirements. Many design changes recommended in the preliminary review were already incorporated, resulting in fewer design deficiencies when reviewed by NUREG 0700 guidance. However, some items identified in the preliminary review had not been incorporated into the design. To assure problems were not overlooked, those items were incorporated in the review as human engineering discrepancies (HEDs) for review and resolution.

18.3.2 OBJECTIVES OF THE CRDR

The objective of the CRDR was to determine the extent to which the VEGP control room provides the operators with sufficient information to complete their required functions and task responsibilities efficiently under emergency conditions. The review also determined the human engineering suitability of the designs of the instrumentation and equipment in the VEGP control room.

To ensure that the CRDR fulfilled its stated purpose, several specific objectives were identified and met during the review. The following specific objectives were defined for the CRDR:

- To perform a control room survey that compares the existing control room with accepted human engineering criteria.
- To review relevant plant operational experience using appropriate documentation and operator questionnaires.
- To determine the information and control requirements of control room operator tasks during emergency conditions.
- To identify human engineering discrepancies.
- To evaluate the extent and importance of identified discrepancies.
- To formulate and implement solutions for significant discrepancies.
- To ensure that the proposed solutions do, in fact, eliminate or mitigate the discrepancies for which they are formulated without creating new discrepancies.
- To verify that implemented solutions eliminate or mitigate identified discrepancies.

18.3.3 CONTROL ROOM DESIGN REVIEW PROCESS

This section describes the process that was used to accomplish the objectives of the CRDR.

18.3.3.1 Preliminary CRDR Status Evaluation

Recognizing that the 1982 control room design review would provide valuable input to the current control room design review, each of the identified deficiencies/discrepancies was reviewed by the use of the following checklist:

- Action reflects the recommended intent and resolves the HED.
- Action reflects the recommended intent but does not resolve the HED.
- Action does not reflect the recommended intent.
- Actions taken create other HED(s).
- No action taken, deficiency remains.
- No action taken, deficiency resolved by resolution of HED.
- New HED(s) generated.
- Other (explain).

18.3.3.2 Operating Experience Review

The Vogtle Electric Generating Plant was under construction with no operating history, and the onsite experience of operational personnel and data from plant operating documents provided little information for the CRDR. Accordingly, the operating experience review focused primarily on industry experience at similar plants and considered the experience gained from the VEGP plant-specific simulator.

Two separate steps were involved in reviewing operating experience. The first was to review available and applicable historical documentation pertaining to plant-specific and generic occurrences. The second step was to survey operating personnel. Operating personnel surveys (operator questionnaire) identified specific problem areas in the VEGP control room that may occur during normal and emergency operation. The operator questionnaire was extracted from the Control Room Design Review Survey Development Guideline (INPO 83-014).

18.3.3.3 Problem Reports

An open ended control room design problem form was sent to all control room and simulator personnel. Copies were made available in the control room and the simulator. This allowed any problem noted to be promptly reported. These problem reports were evaluated for possible HEDs by the review team leader. In many cases, the problem reports identified items already documented by surveys or prior problem reports.

18.3.3.4 Control Room Surveys

Surveys of the existing VEGP control room were conducted during the CRDR. The purpose of the surveys was to compare the design features of the existing control room with applicable human engineering design guidelines. The surveys were conducted by the CRDR review team. The survey team used questionnaires, checklists, and surveys to compile information regarding the as-built characteristics of the VEGP control room.

Eleven separate surveys were completed during the CRDR survey activity. Some of the surveys consisted simply of recording (or determining) control room conventions, such as color usage and instrument arrangement. In general, HEDs were not written during the convention surveys. Instead, the information obtained was used in other CRDR activities to determine where particular instruments or systems departed from the overall convention.

Other surveys measured certain physical quantities, such as illumination and sound level, and compared these measurements to acceptable, or preferred, human engineering standards for such quantities. HEDs were written for characteristics that fell outside the acceptable band.

The individual surveys were:

- General design convention survey (NUTAC 83-042).
- Design convention survey for repetitive groupings (NUTAC 83-042).
- Lighting survey (NUREG 0700 Section 6.1.5).
- Noise survey (NUREG 0700 Appendix E).
- Anthropometric survey (NUTAC 83-042 and NUREG 0700 Section 6.1.2).
- Annunciator survey (NUREG 0700 Section 6.3).
- Communication survey (NUREG 0700 Section 6.2).
- Abbreviation and acronym survey (NUTAC 83-042).
- Color coding survey (NUTAC 83-042).
- Control room computer survey (NUREG 0700 Section 6.7).
- NUTAC 83-042 appendices B-H survey (using applicable NUREG 0700 guidelines).

18.3.3.5 Task Analysis

The operating experience review and the control room survey identified as HEDs those control room characteristics that had caused, or nearly caused, problems during normal operation and simulator exercises and those characteristics that did not conform to certain human engineering design criteria. The task analysis identified the tasks that operators must perform during emergency operation and determined whether the instrumentation, controls, and equipment were available and suitable to perform those tasks. In addition to determining the availability of suitable instrumentation, controls, and equipment, the task analysis validated that the

emergency tasks identified could be performed in real time under simulated emergency conditions in the VEGP control room.

The task analysis used as its basis the Emergency Response Guidelines (ERGs) developed by the Westinghouse Owners Group (WOG).

18.3.3.6 Instrumentation and Control Characteristics Review

In response to the NRC in-progress audit concerns that the task analysis did not identify instrument and control (I&C) needs as compared to the I&C control room inventory, an independent review of the VEGP control room inventory, instrumentation and control characteristics was conducted. The instrumentation and control characteristics review program identified the instrumentation, controls, and characteristics necessary for proper operator response to emergency transients. This review program first identified generic characteristics based on the Westinghouse Owners Group high-pressure reference plant design, followed by the identification of plant-specific deviations. The characteristics of the installed control room instrumentation were justified with the development of or reference to appropriate generic or plant-specific basis documentation.

18.3.3.6.1 Identification of Required Instrumentation and Controls

The Emergency Response Guidelines and the ERG background documents were reviewed to identify:

- All operator tasks necessary to support the operator functions.
- Operator information and control needs necessary to support the operator functions and major actions.
- Plant systems necessary to provide information and control needs.
- Plant instrumentation and controls necessary to provide information and control needs.

For the required plant instrumentation and controls identified above, characteristics were determined based on the information and control needs. The characteristics for the instrumentation included the following:

- Setpoints.
- Units.
- Range.
- Resolution.
- Type of display.

Characteristics for controls included the following:

- Positions.

- Type of control (e.g., variable).

From the information gathered in the characteristics review, a required characteristics justification table was developed for required instrumentation. This table identified operator action categories and associated operator information needs, criteria, and characteristics. The basis for each action category or information need was described or a reference to other documentation was provided.

Following identification of the required characteristics, the VEGP specifics were identified. The VEGP-specific characteristics consisted of applicable generic characteristics and plant-specific deviations. To identify VEGP-specific instrumentation and controls and their associated required characteristics, the Vogtle Emergency Operating Procedures were reviewed to identify deviations from the Emergency Response Guidelines. These VEGP-specific characteristics were then entered in the required characteristics justification tables.

Generic and VEGP-specific required characteristics were reviewed and the limiting required characteristics were summarized in characteristics requirements tables.

18.3.3.6.2 Verification

Verification of the installed control board instrumentation with respect to the above defined required instrumentation and control characteristics was performed. The CRDR team developed an inventory of VEGP control board instrumentation and controls. The CRDR team then compared the inventory to the instrumentation and control characteristics requirements tables for consistency. No HEDs (not already identified in the original task analysis) were identified from this review. This verification provided assurance that the operator did in fact have the required instrumentation and controls assumed in the Westinghouse Owner's Group transient analysis for response to emergency transients.

18.3.4 EVALUATION OF HUMAN ENGINEERING DISCREPANCIES

18.3.4.1 Objectives of Evaluation Process

The objectives of this phase of the CRDR were:

- Evaluate the significance of the HEDs identified in the previous phases of the CRDR.
- Where HEDs were found to be of minor significance, describe the technical and operational basis for such a finding.
- Where the HEDs were found to be significant, formulate changes to the control room design, procedures, operator training, or any combination thereof to mitigate those HEDs.

18.3.4.2 Evaluation Criteria

Human engineering discrepancies found during the control room surveys, the operating experience review, and the task analysis were evaluated by the review team for their potential to adversely affect normal and emergency operation. A categorization scheme was used that

required each HED to be assessed by the review team and prioritized for resolution. The following five categories were designed to be unique so a consensus could be obtained from the review team as to which category each HED should be assigned.

- Category 1 (Safety Significant) - HEDs that have caused errors in or are judged likely to adversely affect the management of emergency conditions by control room operators.

- Category 2 - HEDs that are known to have caused problems during normal operation.

- Category 3 - HEDs that can be fixed with simple and inexpensive enhancements, so called "paint, tape, and label" (PTL) fixes. This included HEDs that were easy to fix but difficult to assess as to the effect on emergency operation.

- Category 4 - These HEDs were judged by the review team as unlikely to affect emergency operation, not documented as causing problems during normal operation, and not easily corrected. However, corrective action was recommended.

- Category 4a - A minor deviation from standards not expected to cause a problem. No corrective action was recommended.

HEDs were initially categorized using a subjective approach that reflected the consensus judgment of the multidisciplinary CRDR team. This HED category review employed a systematic approach based on the assessment process identified in NUREG 0800 and NUREG 0801 (Draft).

HEDs were evaluated with respect to the following attributes and subjected to the algorithm presented in figure 18.3-1 to determine their significance and effects on plant safety.

Item 1 HEDs experienced (EOP validation) or assessed (surveys or checklists) as having a high probability of contributing to operator error.

Item 2 HEDs associated with engineered safety features (ESF) systems.

Item 3 HEDs that could result in unsafe operation or violation of the Technical Specifications.

Item 4 HEDs identified through the operating experience review or actual problems identified in the operator questionnaire.

Item 5 HEDs determined to be easily correctable with paint, tape, labels, engraving changes, or work space environment improvements.

Item 6 HEDs determined to contribute to the operator mental work load (cumulative effects) resulting in fatigue, confusion, or discomfort.

18.3.4.3 Resolution of Human Engineering Discrepancies

18.3.4.3.1 Approach to Correction

The correction of human engineering discrepancies was generally based on the following preferred order of approaches:

A. Guideline compliance - The first preference for correction of HEDs was to modify the control room to comply with the guidelines of NUREG 0700. This approach was used for labeling, procedural, and support equipment HEDs. Control board arrangement HEDs used this approach when consistent with regulatory (train separation) and physical (panel space) limitations.

B. Compensatory measures - When conflicting regulatory or physical constraints prevented a straightforward change to achieve guideline compliance, changes were used which eliminated or reduced the impact of an HED on the operators.

18.3.4.3.2 Engineering Consensus

The development and selection of corrections to HEDs was accomplished by group meetings of the detailed CRDR team. Alternate approaches, benefits, and costs were discussed, and an engineering consensus was arrived at on the recommended approach to correct each HED. The diverse backgrounds of team members were intended to provide input from all disciplines on the resolution of HEDs. The procedures in use called for final recording of the board recommendations. Brief summaries of the resolutions and supporting comments were reported in the Detailed CRDR Report of June 10, 1986.

18.3.4.3.3 Cost-Benefit Analysis

The engineering consensus approach considered costs vs. benefits but did not perform a detailed, documented cost-benefit analysis. The emphasis was on correction of the HEDs to enhance operator performance. The correction of safety significant HEDs was a commitment without regard to cost. In all cases the team members sought to develop the most practical solution balancing constructability, schedule, and cost to achieve the objective of enhanced operator performance.

Some example cost considerations were:

A. Labeling and procedure changes were always implemented to achieve compliance with NUREG 0700. These were typically less than $10,000-projects.

B. Panel rearrangements generally less than $100,000 were implemented.

C. A major control room layout modification of $300,000 was implemented.

D. Major control panel replacements which would cost several million dollars were not recommended. HEDs involving such changes for exact compliance with NUREG 0700 guidelines were addressed with alternate solutions.

18.3.5 COORDINATION WITH OTHER ACTIVITIES

The CRDR process was coordinated with other post-TMI activities in several ways. These activities included the following:

- Upgrading Emergency Operating Procedures.
- Detailed control room design review.
- Post-accident monitoring system (section 7.5).

- Safety parameter display system (subsection 9.5.10).
- Upgrading emergency response facilities.

The mechanism for coordination of control room improvements with other programs was the Emergency Operating Procedures validation program.

The coordination was achieved by: (1) having several members of the detailed CRDR team serve as members of the EOP observation teams during the EOP validation exercises, (2) the task analysis was a common basis for both the detailed CRDR and EOP programs, and (3) the findings or discrepancies identified during the validation exercises related to man and machine interface were processed via the detailed CRDR HED assessment process.

In addition, control room modifications that resulted from the Emergency Operating Procedures validation program were addressed in the detailed CRDR verification plan to ensure design improvements provided the necessary correction.

18.3.5.1 Emergency Operating Procedures

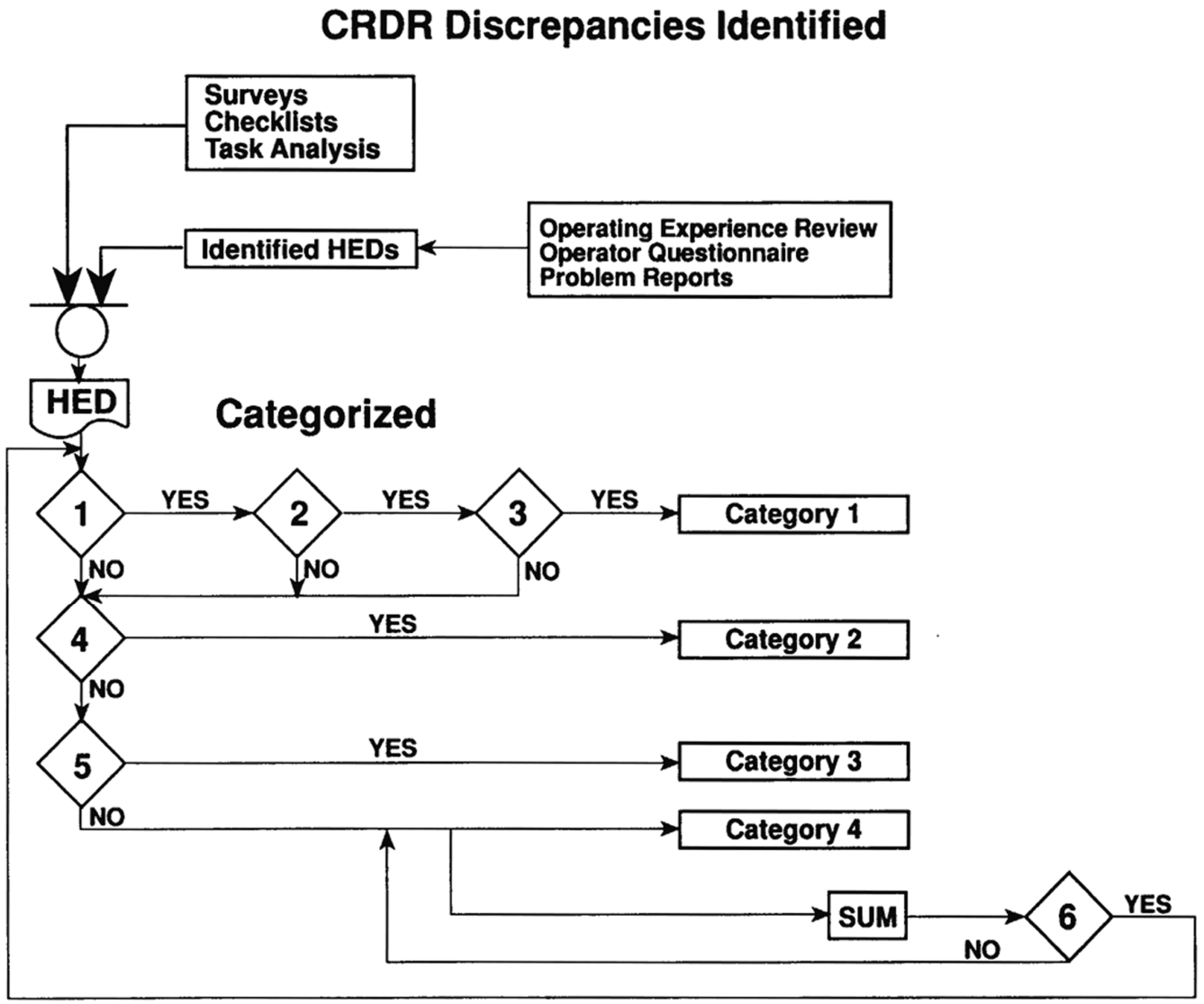
The task analysis portion of the detailed CRDR used the Westinghouse ERGs as plant-specific EOPs as its starting point. Thus, the task of upgrading emergency procedures is inherently integrated into the detailed CRDR. The simulator validation of the task analysis was combined with the VEGP EOP validation program.

18.3.5.2 Safety Parameter Display System (SPDS)

GPC used the SPDS during the EOP validation exercise, and evaluated it to NUREG 0700 Section 6.7 (process computer) guidelines. Some HEDs identified during the EOP validation exercise or the computer survey and judged to be significant by the review team were resolved by incorporating certain features into the SPDS and associated displays. In addition, the detailed CRDR team leader was chairman of a control room computer task force and participated in developing displays and board arrangements. This served to incorporate human engineering requirements into the design of the SPDS and further integrate them into the detailed CRDR process. (See the Emergency Plan, section H.4.5.)

18.3.5.3 Regulatory Guide 1.97

The design of Regulatory Guide 1.97 instrumentation was essentially complete at the time of this review. This instrumentation was evaluated in the detailed CRDR survey and task analysis.



REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

HUMAN ENGINEERING DISCREPANCY EVALUATION ALGORITHM

FIGURE 18.3–1 (SHEET 1 OF 2)

Algorithm Attributes

ITEM 1

HEDs experienced or assessed as having a high probability of contributing to operator error.

ITEM 2

HEDs associated with engineered safety features systems.

ITEM 3

HEDs that could result in unsafe operation or violation of the Technical Specifications.

ITEM 4

HEDs identified through the operation experience review or actual problems identified in the operator questionnaire.

ITEM 5

HEDs determined to be easily correctable with paint, tape, labels, engraving changes, or work space environment improvements.

ITEM 6

HEDs determined to contribute to operator mental work load (cumulative effects) resulting in fatigue, confustion, or discomfort.

REV 14 10/07